

Physics, 2018 EC, entrance exam, version 4

Grade 12 · Physics · 2018 E.C.

Time allowed: 120 minutes

Questions: 50

Total marks: 50

Name _____ Grade _____ ID No. _____ Date _____

1. Which of the following statements is FALSE about vector physical quantities? (1)
 - A. Multiplying a vector by a negative number changes both the magnitude and direction of the vector.
 - B. The dot product of two vectors becomes maximum when the two vectors are the same in direction.
 - C. The magnitude of the sum of two vectors is equal to the magnitude of their difference when the two vectors are perpendicular to each other.
 - D. The resultant of two vectors is minimum when the vectors make 45° between them.

2. If two vectors are given as: $A = 3i + 3j$ and $B = 2i + 4j$, which of the following is NOT true about the two vectors? (1)
 - A. The dot product of the two vectors is 18
 - B. The scalar projection of vector A in the direction of vector B is $9/\sqrt{5}$
 - C. The unit vector in the direction of $A + B$ is $(5i+7j)/\sqrt{74}$
 - D. The angle between the two vectors is $\sin^{-1}(3/\sqrt{10})$

3. A force of 80 N is applied on an object at an angle of 37° east of north. What are the east and north components of the force? (1)
 - A. $F_{\text{east}} = 48\text{N}$ and $F_{\text{north}} = 64\text{N}$
 - B. $F_{\text{east}} = 64\text{N}$ and $F_{\text{north}} = 48\text{N}$
 - C. $F_{\text{east}} = F_{\text{north}} = 48\text{N}$
 - D. $F_{\text{east}} = F_{\text{north}} = 64\text{N}$

4. Two vectors A and B are perpendicular to each other. The magnitude of the resultant is $\sqrt{1000}$ units and $A = 3B$. What are the magnitudes of A and B? (1)
 - A. $A = 10$ units and $B = 30$ units
 - B. $A = 30$ units and $B = 10$ units
 - C. $A = 40$ units and $B = 30$ units
 - D. $A = 30$ units and $B = 40$ units

5. The magnitudes of two vectors A and B are 12 units and 5 units, respectively. Which one of the following is NOT the possible value for the magnitude of the resultant vector? (1)
 - A. 20 units
 - B. 17 units
 - C. 13 units
 - D. 7 units

6. A ball is thrown downward from the top of a building with an initial speed of 40 m/s. It strikes the ground after 4s. How high is the building, assuming negligible air resistance? (1)
 - A. 180 m
 - B. 480 m
 - C. 170 m
 - D. 240 m

7. A car moves at a speed of 20 m/s on a banked road with a radius of $40\sqrt{3}$ m. What is the banking angle of the road? (1)
 - A. 45°
 - B. 30°
 - C. 60°
 - D. 25°

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8. If the displacement of a particle is expressed as $x(t) = 2t^2 + 10t - 4$, where x is in meter and t is in second, what are the instantaneous velocity and instantaneous acceleration at $t = 2$ sec? (1)
- A. $v_{ins} = 18\text{m/s}$ and $a_{ins} = 18\text{m/s}^2$ B. $v_{ins} = 18\text{m/s}$ and $a_{ins} = 4\text{m/s}^2$
C. $v_{ins} = 4\text{m/s}$ and $a_{ins} = 18\text{m/s}^2$ D. $v_{ins} = 4\text{m/s}$ and $a_{ins} = 4\text{m/s}^2$
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9. Which of the following is FALSE statement in projectile motion? (1)
- A. The vertical component of the velocity changes with time due to the action of gravity
B. The horizontal component of the velocity is constant due to the absence of net force in that direction
C. The direction of acceleration is always downward throughout the projectile's motion
D. The vertical and the horizontal components of projectile's motion are dependent on each other
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10. A football player kicks a ball with a velocity of $u = 10\sqrt{10}$ m/s at an angle of $\theta = 37^\circ$ above the horizontal. What are the range and the maximum height of the ball, respectively? (1)
- A. 24m and 36m B. 96m and 18m
C. 36m and 48m D. 18m and 48m
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11. If the value of the gravitational acceleration on the surface of the earth is g , then what will be its value at a distance of $h = R_E$ above the surface of the Earth? (1)
- A. $4g$ B. $g/2$
C. $g/4$ D. $g/8$
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12. A horizontally mounted wheel of radius R is initially at rest, and then begins to accelerate constantly until it has reached an angular velocity ω after 5 complete revolutions. What was the angular acceleration of the wheel? (1)
- A. $\omega^2/(20\pi)$ B. $\omega^2/(10\pi R)$
C. $\omega^2/(5\pi R)$ D. $\omega^2/(10\pi)$
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13. What happens to the gravitational force between two masses if each mass is doubled and the separation distance between them is halved? (1)
- A. Increases by a factor of 16 B. Decreases by a factor of 16
C. Increases by a factor of 8 D. Decreases by a factor of 8
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14. If 20% of an unknown object is above the surface of water, what is the density of the unknown object? (1)
- A. 800 kg/m^3 B. 200 kg/m^3
C. 400 kg/m^3 D. 600 kg/m^3
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15. A U-tube manometer is used to measure the pressure of a gas. One end of the tube is connected to the gas, and the other is open to the atmosphere. If the liquid level is higher on the gas side, what does this indicate about the gas pressure? (1)
- A. The gas pressure is equal to atmospheric pressure.
B. The gas pressure is greater than atmospheric pressure.
C. The gas pressure is less than atmospheric pressure.
D. Additional information is needed to draw a conclusion.

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16. A skater spins with her arms extended. When she pulls her arms closer to her body, which of the following occurs? (1)
- A. Her angular velocity decreases and her angular momentum remains constant.
 - B. Her angular velocity increases and her angular momentum increases.
 - C. Her angular velocity increases and her angular momentum remains constant.
 - D. Both angular velocity and angular momentum remain constant.
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17. Suppose that water flows from a pipe with a diameter of 2cm into another pipe of diameter 4cm. If the speed of water in the first pipe is 20 m/s, what is the speed of water in the second pipe? (1)
- A. 10 m/s
 - B. 40 m/s
 - C. 80 m/s
 - D. 5 m/s
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18. The Center of curvature of a concave mirror is 40 cm. If an object is placed 10 cm in front of it, the image will be: (1)
- A. Virtual, upright, and magnified
 - B. Real, upright, and diminished
 - C. Real, inverted, and magnified
 - D. Virtual, inverted, and diminished
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19. What is the main difference between primary and secondary waves in seismic waves? (1)
- A. Primary waves are transverse waves whereas secondary waves are longitudinal waves.
 - B. Secondary waves are transverse waves whereas primary waves are longitudinal waves.
 - C. Both transverse waves and primary waves are longitudinal waves
 - D. Both transverse waves and primary waves are transverse waves
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20. A 60 years-old patient presents with a persistent headaches and a suspected brain tumor. Which imaging modality is most appropriate for providing a detailed image of the brain? (1)
- A. X-ray
 - B. CT-scan
 - C. MRI
 - D. Ultrasound
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21. Several containers of different sizes and shapes are all filled with the same liquid to the same depth. Then: (1)
- A. The force of the liquid on the bottom of each container is the same
 - B. The least pressure is at the bottom of the container with the largest bottom area
 - C. The greatest pressure is at the bottom of the container with the largest bottom area
 - D. The pressure on the bottom of each container is the same
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22. A 500 kg car moving at 13 km/hr collides and couples with an 800 kg car which is initially at rest. The approximate common final speed of these two cars is (1)
- A. 13 km/h
 - B. 5 km/h
 - C. 2.5 km/h
 - D. 10 km/h
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23. Two charges are located on the line shown in the figure below, in which the charge at point I is $+3q$ and the charge at point III is $+2q$. Point II is halfway between points I and III. Other than at infinity, the electric field strength is zero at point on the line in which of the following ranges? (1)
- A. To the left of I
 - B. Between I and II
 - C. Between II and III
 - D. To the right of III
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24. Which of the following formulas correctly expresses the resistivity of a resistor that has resistance R , length L , and cross-sectional area A ? (1)
- A. $\rho = AL/R$
 - B. $\rho = RL/A$
 - C. $\rho = RA/L$
 - D. $\rho = RLA$

25. A battery with an e.m.f of 12 V delivers 10 V to an external circuit when the current in the circuit is 1 A. What is the internal resistance of the battery? (1)

A. $0.5\ \Omega$ B. $1\ \Omega$
C. $2\ \Omega$ D. 0

26. A parallel plate capacitor stores $100\ \mu\text{J}$ of energy when charged to an electric potential difference of 20 V. If the same capacitor is charged again to a potential difference of only 10 V, what will be the energy stored in the capacitor? (1)

A. $200\ \mu\text{J}$ B. $100\ \mu\text{J}$
C. $50\ \mu\text{J}$ D. $25\ \mu\text{J}$

27. What is the electric field at a distance of 2 cm from the center of an object with a charge of 2nC ? (1)

A. $4,500\ \text{N/C}$ B. $9,000\ \text{N/C}$
C. $45,000\ \text{N/C}$ D. $90,000\ \text{N/C}$

28. A metal sphere with a charge of $+2Q$ comes into contact with a metal sphere of identical size that has a charge of $-4Q$. The spheres are then separated. What are the charges on the spheres after they are separated? (1)

A. Both spheres have zero net charge. B. Each sphere has a charge of $-Q$.
C. Each sphere has a charge of $-2Q$. D. Each sphere retains its original charge.

29. A 2,000 kg automobile accelerates from rest to a speed of 40 m/s in 20 s. What is the average power, in kilowatts, produced by the automobile during this time interval? (1)

A. 10 kW B. 20 kW
C. 40 kW D. 80 kW

30. Which one of the following is the CORRECT relationship in measuring the distance of astronomical object by using its brightness? (1)

A. Apparent brightness $\propto$ true brightness/distance
B. Apparent brightness $\propto$ true brightness/distance²
C. Apparent brightness $\propto$ (true brightness)(distance²)
D. Apparent brightness $\propto$ distance²/true brightness

31. A bar magnet is divided in two pieces. Which of the following statements is true about the force between the broken pieces if they face each other with a small separation? (1)

A. There is an electric repulsive force between the broken pieces
B. There is an electric attractive force between the broken pieces
C. There is a magnetic repulsive force between the broken pieces
D. There is a magnetic attractive force between the broken pieces

32. A magnetic field with a flux density of $2 \times 10^{-2}\ \text{T}$ is directed at an angle of 30° to a surface area of $1 \times 10^{-1}\ \text{m}^2$. What is the magnetic flux through the surface? (1)

A. $2 \times 10^{-3}\ \text{Wb}$ B. $3 \times 10^{-3}\ \text{Wb}$
C. $4 \times 10^{-3}\ \text{Wb}$ D. $1 \times 10^{-3}\ \text{Wb}$

33. A transformer has 200 turns in its primary coil and 800 turns in its secondary coil. If the input voltage is 120V, what is the output voltage? (1)

A. 30V B. 120V
C. 480V D. 960V

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- 34.** Which one of the following is FALSE about waves? (1)
- A. Mechanical waves require material medium for their propagation
 - B. Electromagnetic waves can travel through vacuum
 - C. In transverse waves, vibration of particles/fields is parallel to wave motion
 - D. Sound wave and seismic waves are examples of mechanical waves.
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- 35.** An object sinks in the liquid when (1)
- A. The weight of the object is less than the buoyant force of the liquid
 - B. The weight of the object is equal to the buoyant force of the liquid
 - C. The weight of the object is greater than the buoyant force of the liquid
 - D. The buoyant force is equal to the weight of the displaced liquid
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- 36.** Two small hollow metal spheres hung on insulating threads attract one another as shown. If a positively charged rod attracts metal sphere A, which one of the following is TRUE about the charge on the metal spheres? (1)
- A. Metal sphere A has positive charge
 - B. Metal sphere B has positive charge
 - C. Both metal spheres have positive charge
 - D. Both metal spheres have negative charge
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- 37.** How does specific heat capacity affect the rate at which a substance heats up or cools down? (1)
- A. A substance with high specific heat capacity heats up or cools down quickly.
 - B. A substance with low specific heat capacity heats up or cools down slowly.
 - C. A substance with high specific heat capacity heats up or cools down slowly.
 - D. Additional information is needed to draw a conclusion.
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- 38.** A rectangular iron sheet has dimensions of 4m by 2m. If the temperature increases from 10°C to 60°C, what is the new area of the sheet? ($\alpha_{\text{iron}} = 1.5 \times 10^{-5}/^{\circ}\text{C}$) (1)
- A. 8.012 m²
 - B. 7.988 m²
 - C. 1.2×10^{-2} m²
 - D. 4.006 m²
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- 39.** Three masses, 3 kg, 2 kg, and 5kg, are placed at (1, 2) m, (2, 1) m and (-3, -4) m, respectively. Where is the center of mass of the system? (1)
- A. (0.8, 1.2) m
 - B. (-0.8, -1.2) m
 - C. (-0.8, 1.2) m
 - D. (0.8, -1.2) m
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- 40.** Which one of the following is TRUE about semiconductors? (1)
- A. Pure semiconductors act like insulators at high temperature.
 - B. P-type semiconductors are produced by adding pentavalent elements to pure semiconductors.
 - C. Adding impurities increases the electrical conductivity of semiconductor materials.
 - D. The conductivity of pure semiconductors is inversely proportional to temperature.
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- 41.** What is the value of the emitter current for a common emitter transistor with a base current of 40 μA and a current gain of 150? (1)
- A. 6 mA
 - B. 8 mA
 - C. 5.96 mA
 - D. 6.04 mA

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- 42.** One of the following is FALSE in the conservation of mechanical energy: (1)
- A. Energy is neither created nor destroyed but it can be transformed from one type to another
 - B. When kinetic energy increases, potential energy decreases
 - C. When kinetic energy decreases, potential energy increases
 - D. Both kinetic energy and potential energy can increase at the same time
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- 43.** A mass of 1kg moving with a speed of 2 m/s on a horizontal smooth surface, collides with a nearly weightless spring of force constant $k = 100 \text{ N/m}$. The maximum compression of the spring would be (1)
- A. 20cm
 - B. 40cm
 - C. 10cm
 - D. 30cm
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- 44.** A ball of mass m with speed v strikes a wall at an angle θ with the normal, as shown in the diagram below. It then rebounds with the same speed and at the same angle. The impulse delivered by the ball to the wall is (1)
- A. $mv \sin\theta$
 - B. $mv \cos\theta$
 - C. $2mv \sin\theta$
 - D. $2mv \cos\theta$
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- 45.** A car of mass m is driving around a circular track of radius r at a constant velocity of v . The centripetal force acting on the car is FC . If the car's velocity is doubled, what is the new centripetal force required for the car to drive on the circular track? (1)
- A. $FC/2$
 - B. $4FC$
 - C. $FC/4$
 - D. $2FC$
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- 46.** What happens to the kinetic energy of a body if its speed increases by a factor of 3? (1)
- A. Decreases by a factor of 9
 - B. Increases by a factor of 9
 - C. Increases by a factor of 3
 - D. Decreases by a factor of 3
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- 47.** Which one of the following is TRUE statement about friction force? (1)
- A. Static friction is less than kinetic friction
 - B. Static friction is greater than kinetic friction
 - C. Static friction is equal to kinetic friction
 - D. Rolling friction is greater than sliding friction
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- 48.** Which one does NOT affect the magnitude of induced electromotive force in the coil of wire? (1)
- A. The rate of change of magnetic flux
 - B. The number of turns in the coil
 - C. The area of the coil
 - D. The resistance of the coil
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- 49.** Which electrical component generates electric current when light strikes on it? (1)
- A. Photo diode
 - B. Light emitting diode
 - C. Resistor
 - D. Transistor
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- 50.** A coil with 100 turns is placed in a magnetic field that changes from 0.4T to 0.1T in 2 seconds. If the area of the coil is 0.1 m^2 , what is the maximum induced electromotive force in the coil? (1)
- A. 1.5 V
 - B. 0.75 V
 - C. 1 V
 - D. 0.5 V

— END OF EXAMINATION —